Supplementary Results: Rank-Aware Evaluation of Approximate Nearest Neighbor Graphs for Minimum Spanning Tree Preservation

Mahmood K. M. Almansoori, Andras Meszaros, Miklos Telek

1 Overview

This document contains the complete rank-recall results underlying the analysis in the main paper. For each of the 13 benchmark datasets and each evaluated value of

$$k \in \{5, 10, 15, 20, 25, 30, 35, 40, 45, 50\},$$

we report the rank-recall curve

$$\widetilde{r}(t) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\left[\nu(i,t) \in \widetilde{N}_k(i)\right], \qquad t \in \{1, \ldots, n-1\},$$

as defined in the main paper. Results are provided for both approximate nearest-neighbor methods considered in the paper, PyNNDescent and FAISS HNSW. The figures are organized in two complementary views:

- **Per-dataset overlays** (Section 3): one figure for each dataset, with all evaluated values of k overlaid. This view shows how the rank-recall curve changes with k for a fixed dataset.
- **Per-k overlays** (Section 4): one figure for each value of k, with all 13 datasets overlaid. This view facilitates comparison across datasets at a fixed k.

2 Datasets

Table 1 lists the 13 benchmark datasets used in the experiments. Here, n denotes the number of points after duplicate removal and m denotes the dimensionality. Dataset sources and the corresponding Minimum Spanning Tree (MST) edge-rank statistics are reported in Table 1 of the main paper.

Table 1: Datasets used in the rank-recall sweep, ordered by size.

dataset	n	m
Speech	3,686	400
MissAmerica	6,480	16
Unbalance	6,500	2
MNIST	7,603	100
Shape	28,775	544
House	34,112	3
Shuttle	49,097	9
Audio	54,387	192
F-MNIST	70,000	784
Birch1	100,000	2
CelebA	113,983	39
Europe	168,435	2
Corel	661,158	14

3 Per-dataset overlays

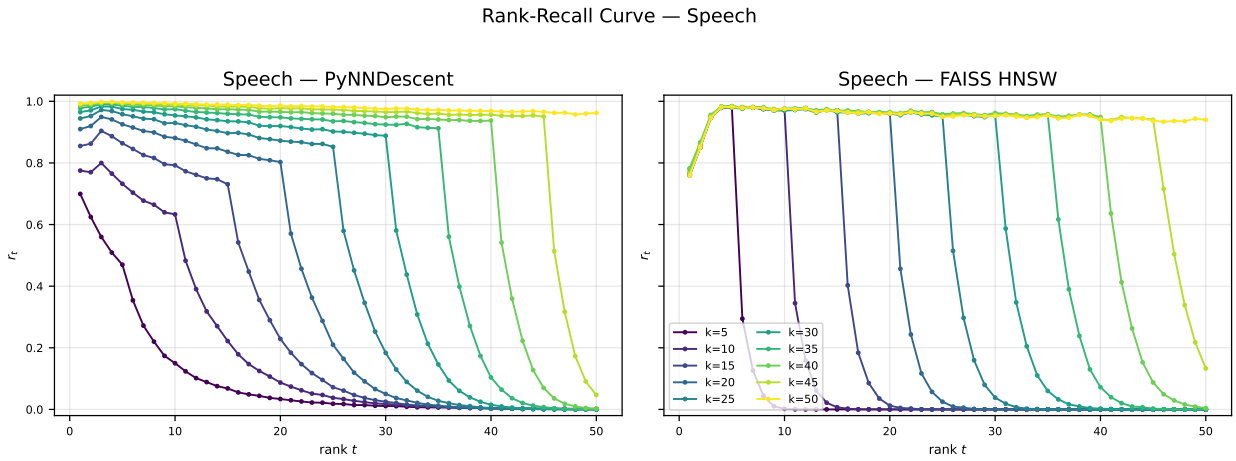


Figure 1: Rank-recall curve $\widetilde{r}(t)$ for Speech. Speech is the only dataset in the sweep for which FAISS HNSW's recall is non-monotone at low ranks (see Section 5 of the main paper, "When does Recall@k exceed MER?").

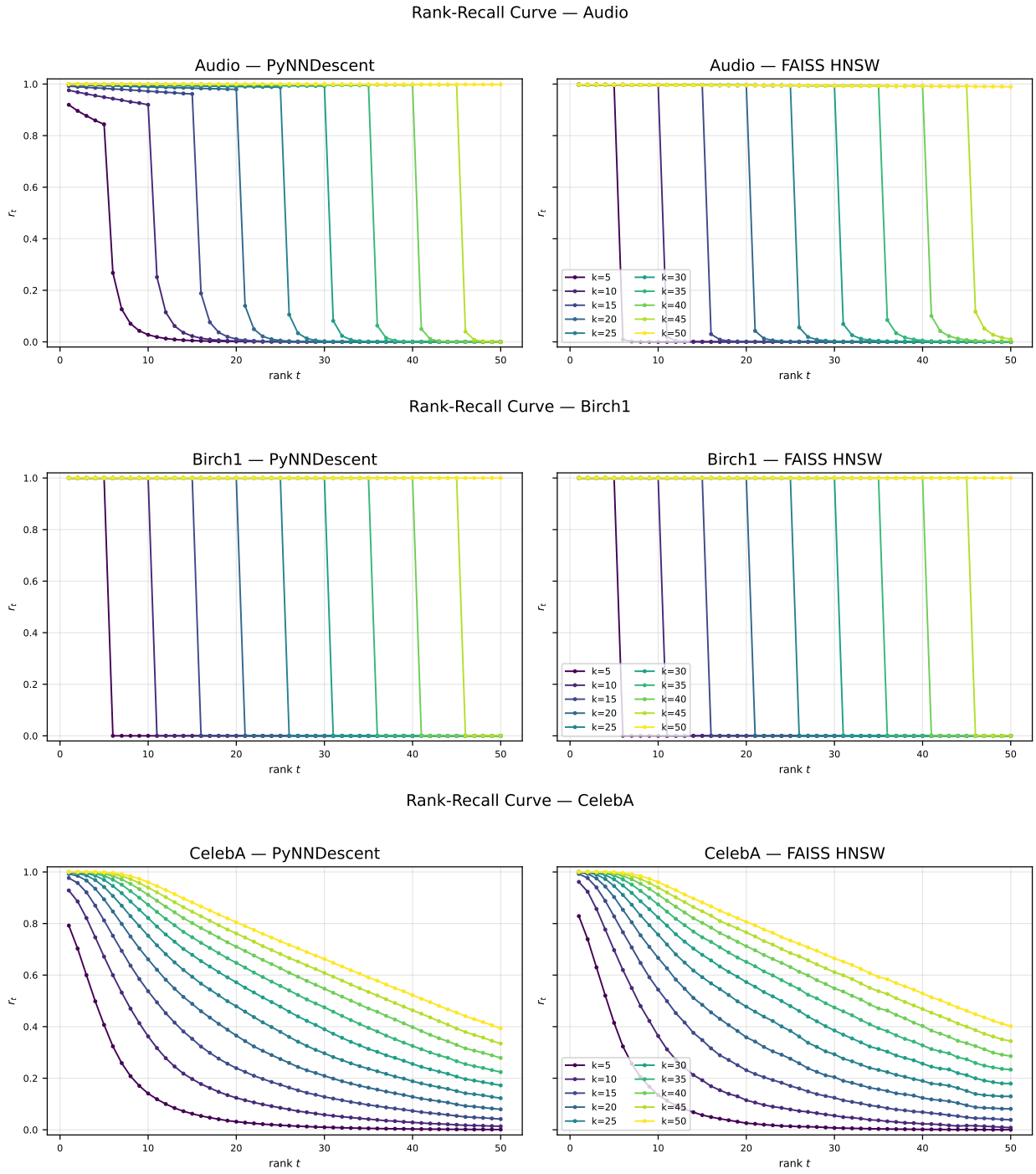


Figure 2: Rank-recall curve $\widetilde{r}(t)$ for Audio, Birch1, CelebA.

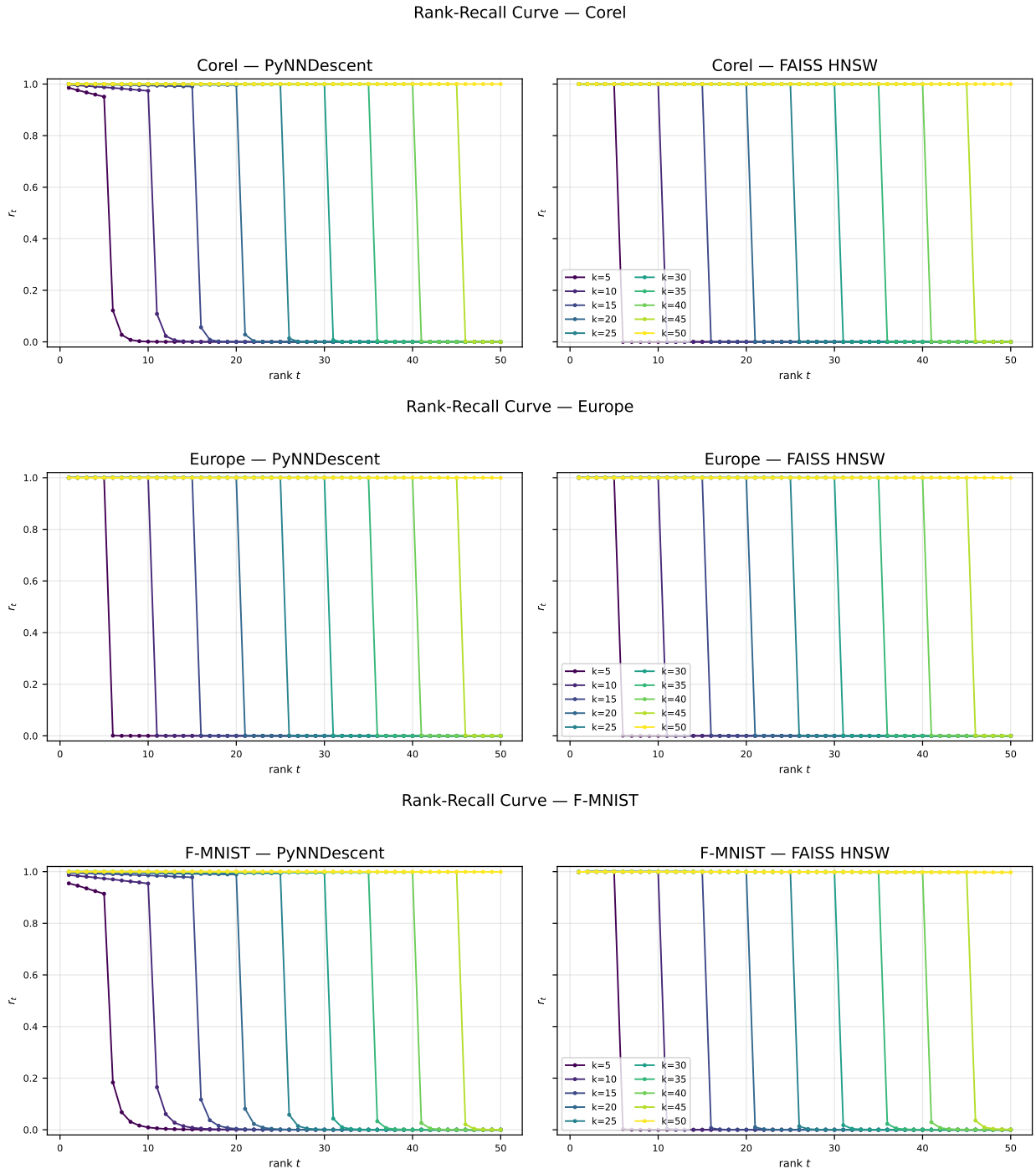


Figure 3: Rank-recall curve $\widetilde{r}(t)$ for Corel, Europe, F-MNIST.

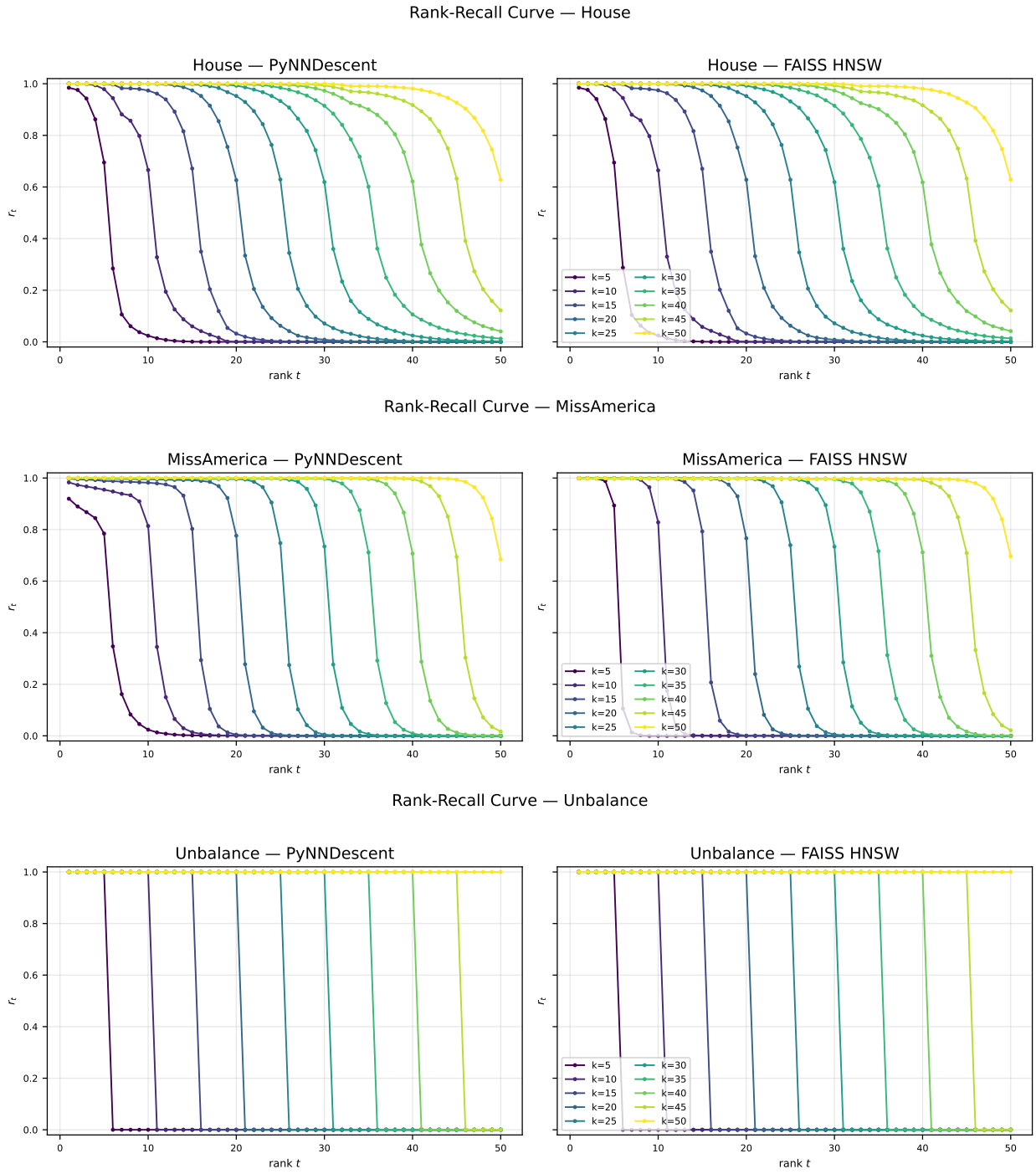


Figure 4: Rank-recall curve $\widetilde{r}(t)$ for House, MissAmerica, and Unbalance.

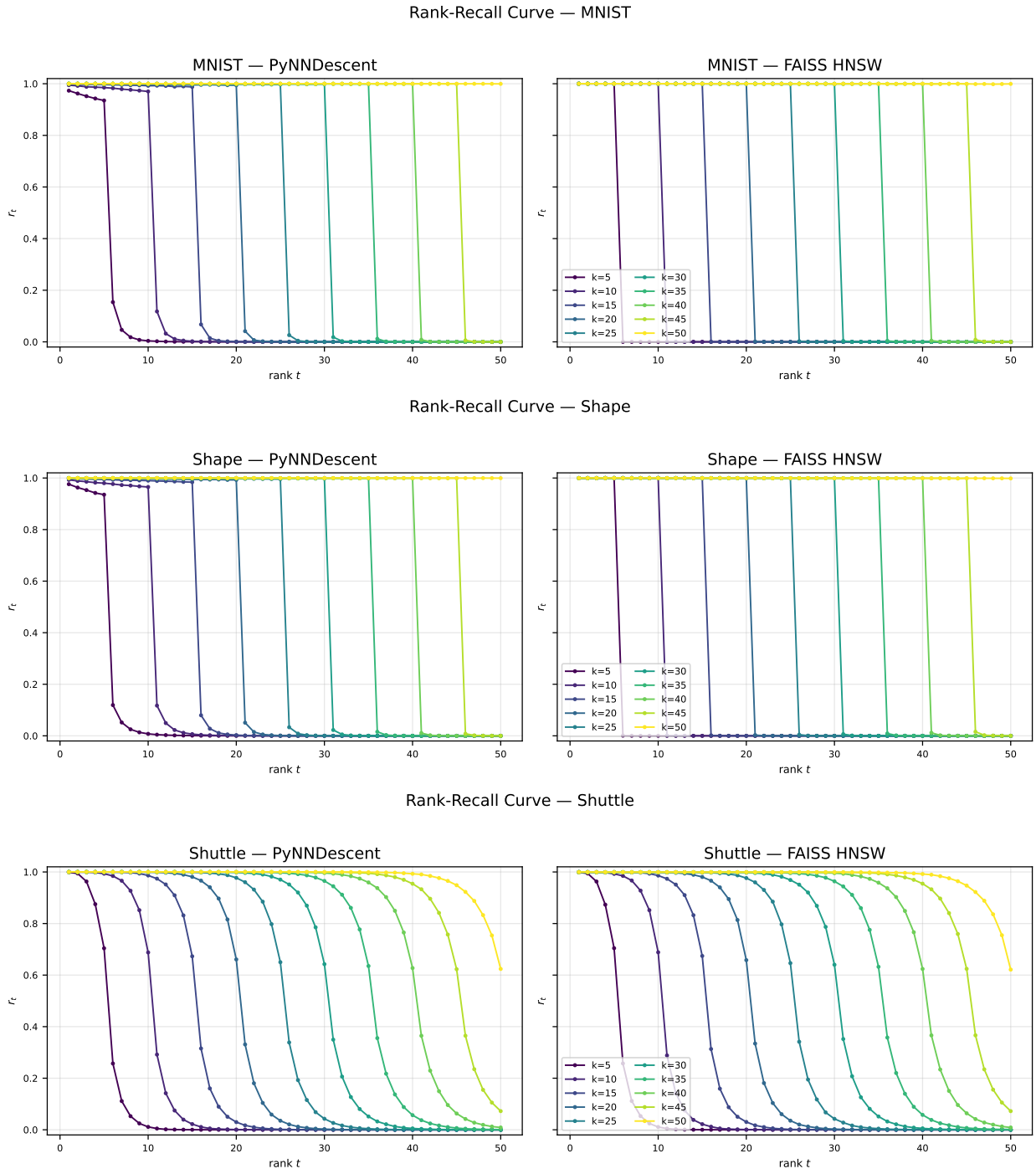


Figure 5: Rank-recall curve $\widetilde{r}(t)$ for MNIST, Shape, Shuttle.

4 Per-k overlays

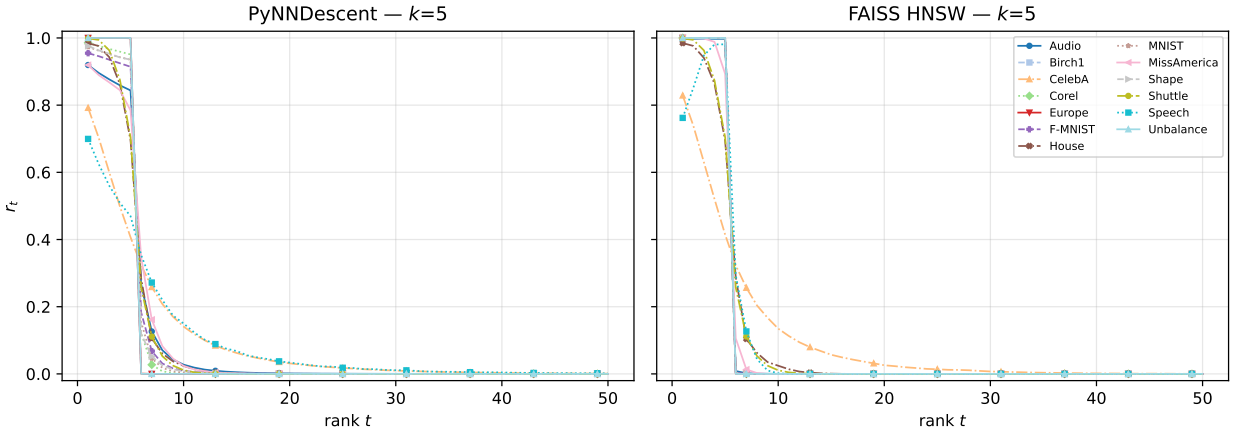


Figure 6: Rank-recall curves $\widetilde{r}(t)$ for $k = 5$ across all 13 datasets. Left: PyNNDescent. Right: FAISS HNSW. The conservation identity gives $\sum_t \widetilde{r}(t) = k$ for every curve; the portion with $t > k$ is the rank-recall tail.

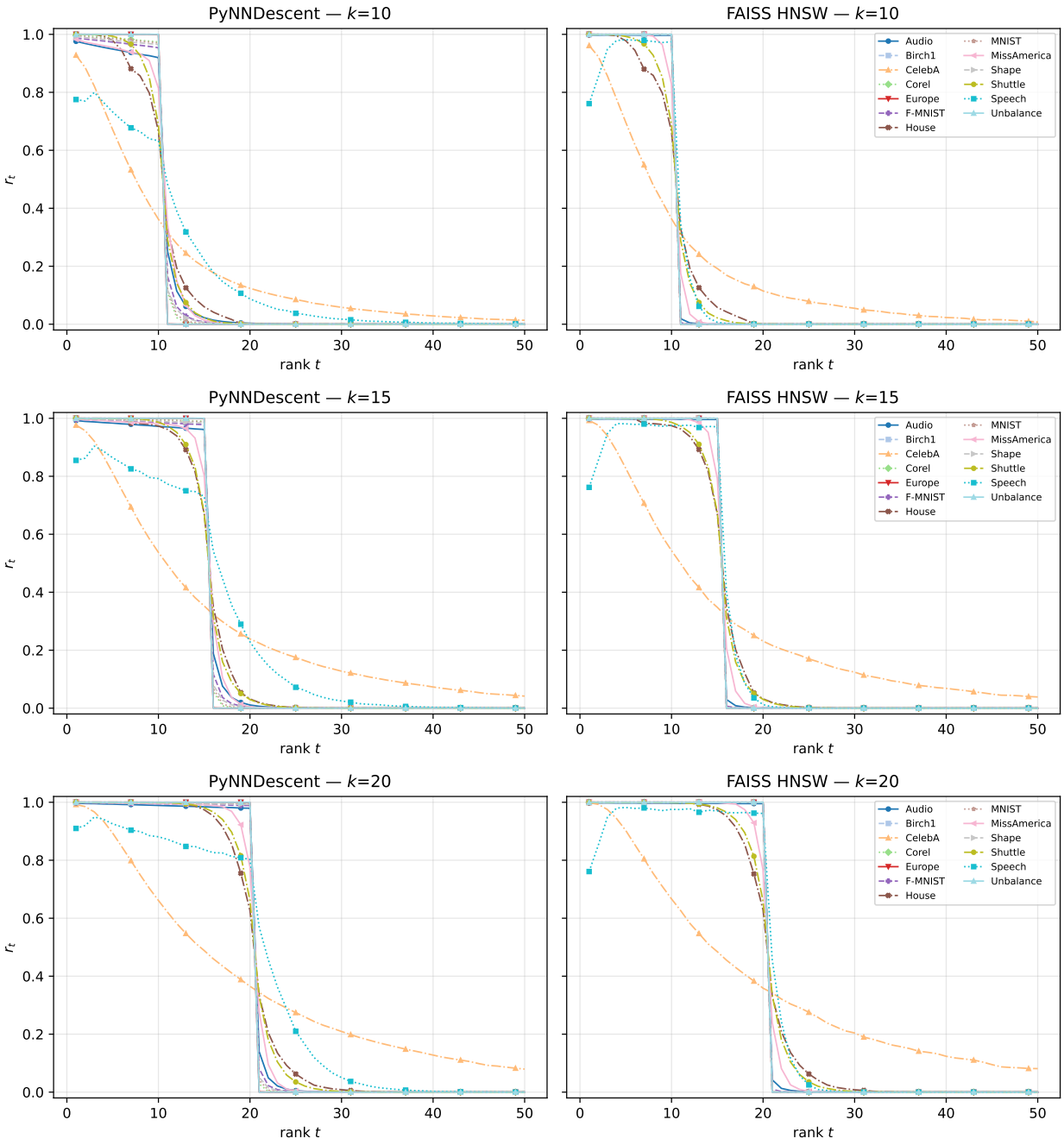


Figure 7: Rank-recall curve $\widetilde{r}(t)$ for $k \in \{10, 15, 20\}$.

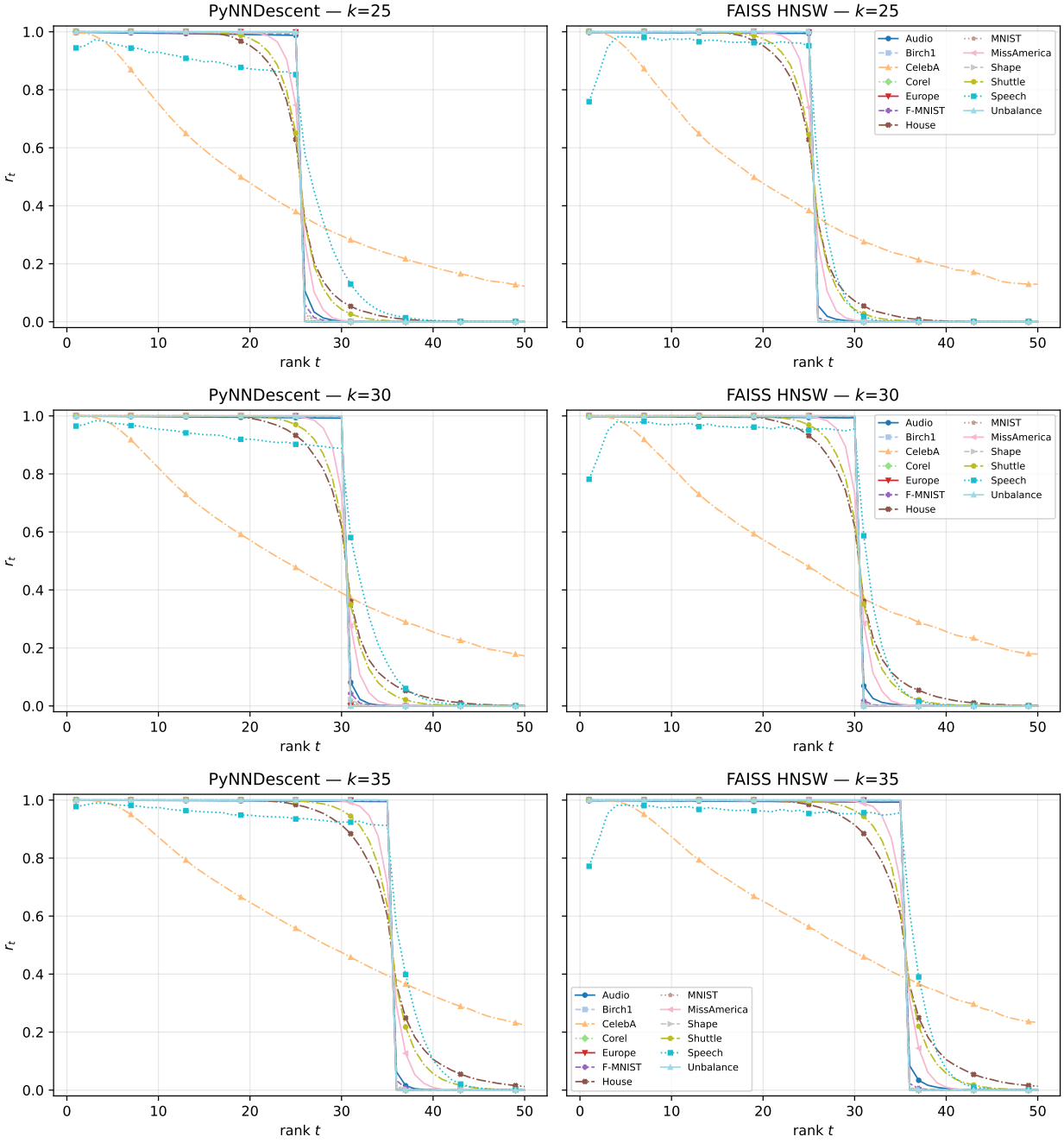


Figure 8: Rank-recall curve $\widetilde{r}(t)$ for $k \in \{25, 30, 35\}$.

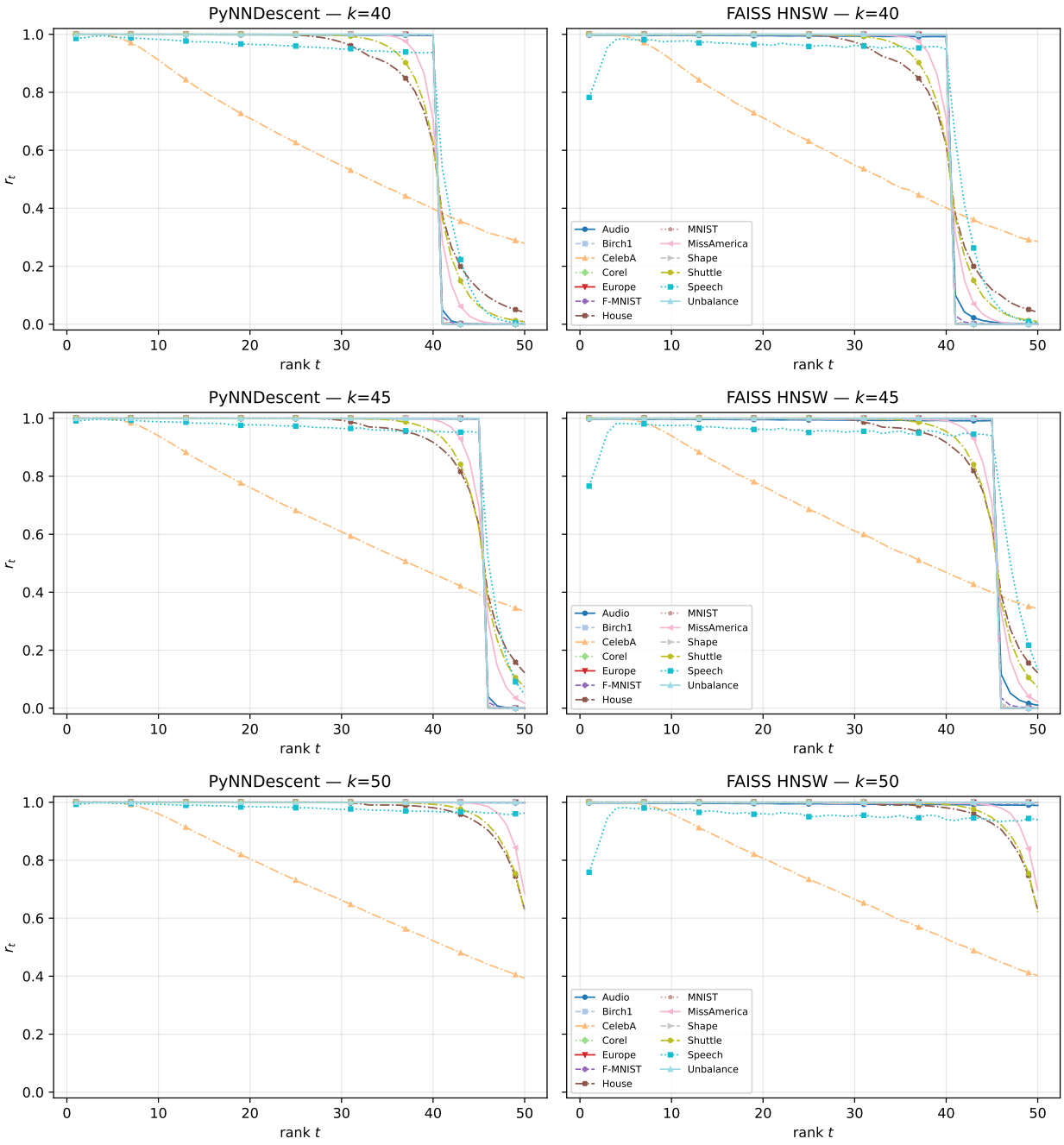


Figure 9: Rank-recall curve $\widetilde{r}(t)$ for $k \in \{40, 45, 50\}$.